

Experiment 24: Diet Recommendation Based on Disease Using Prolog

Aim

To develop a **Prolog program that suggests a suitable diet based on a given disease** using facts, rules, and logical queries.

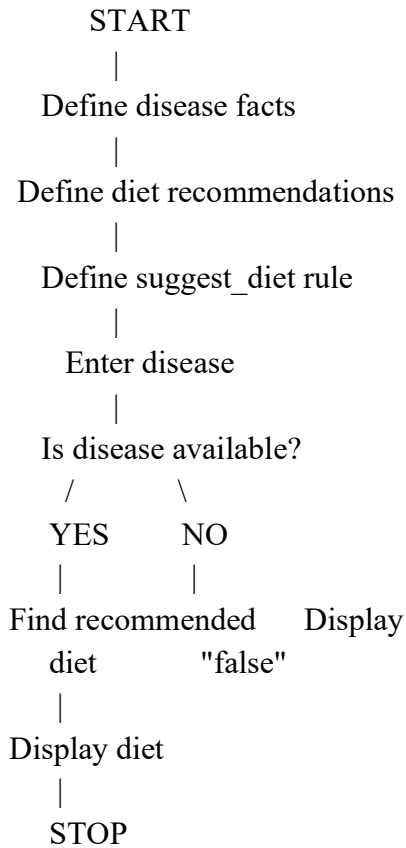
Objective

1. To represent diseases and diet recommendations using Prolog facts.
2. To create rules for connecting diseases with suitable diets.
3. To understand knowledge representation using Prolog.
4. To execute queries and obtain diet recommendations.

Algorithm

1. Start the program.
2. Define facts representing different diseases.
3. Define diet recommendations for each disease.
4. Create a rule `suggest_diet` to connect a disease with its recommended diet.
5. Enter the disease as a query.
6. Prolog searches the knowledge base.
7. If the disease exists, display the corresponding diet recommendation.
8. If the disease does not exist, return false.
9. Stop.

Flowchart



Prolog Code

% Diet Recommendation Based on Disease

% Disease facts

disease(diabetes).

disease(hypertension).

disease(anemia).

disease(obesity).

disease(gastritis).

% Diet recommendations

diet(diabetes, 'Low sugar and high fiber foods').

diet(hypertension, 'Low salt and potassium rich foods').

diet(anemia, 'Iron rich foods such as spinach and legumes').

diet(obesity, 'Low calorie foods, vegetables and fruits').

diet(gastritis, 'Light foods, avoid spicy and oily foods').

```
% Rule to suggest diet
suggest_diet(Disease, Diet) :-
    disease(Disease),
    diet(Disease, Diet).
```

Sample Output

Query 1: Diabetes

?- suggest_diet(diabetes, Diet).

Query 2: Hypertension

?- suggest_diet(hypertension, Diet).

Query 3: Anemia

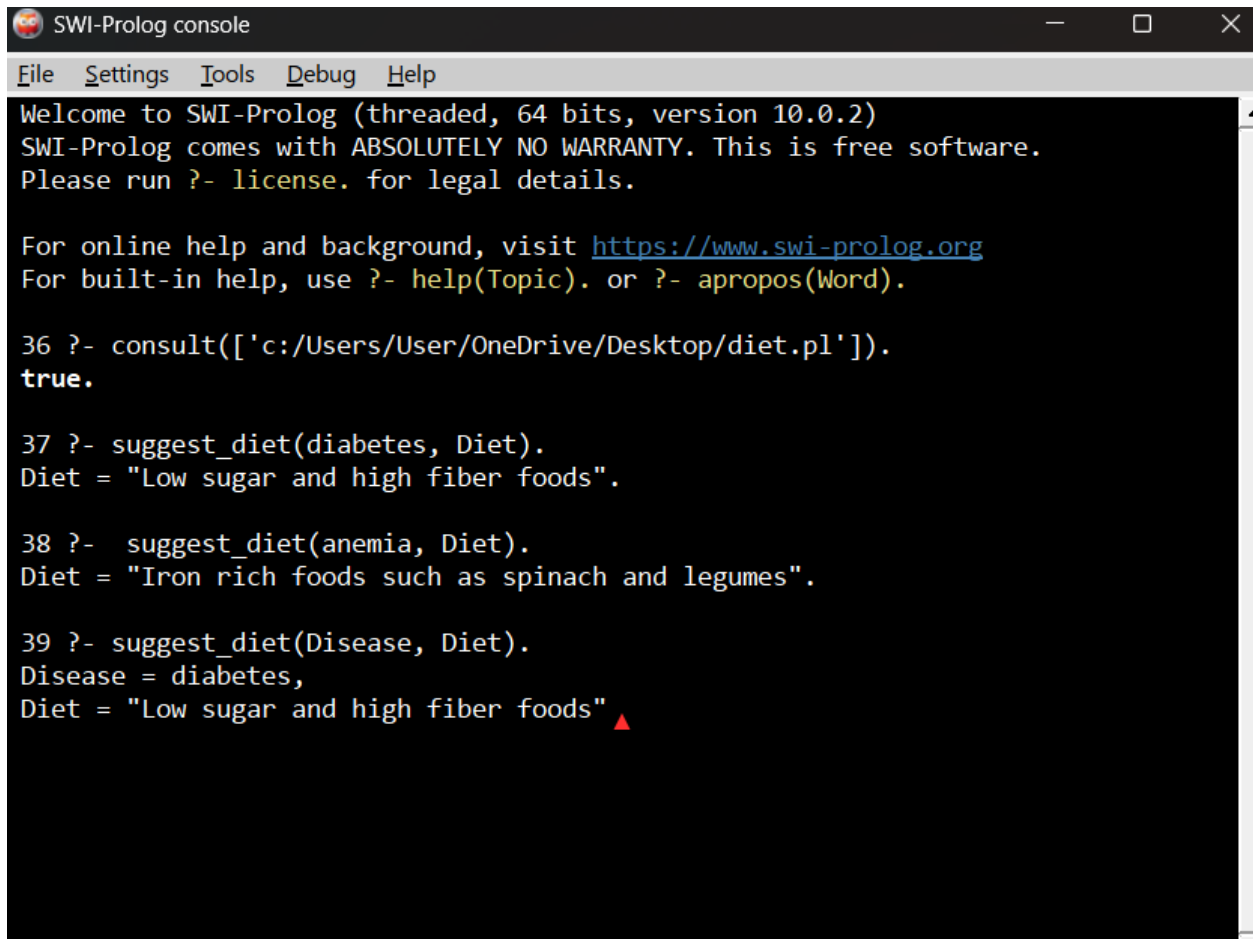
?- suggest_diet(anemia, Diet).

Query 4: Obesity

?- suggest_diet(obesity, Diet).

Query 5: Find diets for all diseases

?- suggest_diet(Disease, Diet).



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

36 ?- consult(['c:/Users/User/OneDrive/Desktop/diet.pl']).
true.

37 ?- suggest_diet(diabetes, Diet).
Diet = "Low sugar and high fiber foods".

38 ?- suggest_diet(anemia, Diet).
Diet = "Iron rich foods such as spinach and legumes".

39 ?- suggest_diet(Disease, Diet).
Disease = diabetes,
Diet = "Low sugar and high fiber foods" ▲
```

Result

The **diet recommendation system** was successfully implemented using Prolog. The program correctly suggests a predefined diet recommendation when a particular disease is given as a query.

Conclusion

Thus, the experiment demonstrates how **Prolog facts and rules can be used to build a simple knowledge-based recommendation system**. The system successfully maps diseases to predefined diet recommendations using logical inference.

Note: This is an academic demonstration only; actual disease-specific diets should be determined by a qualified healthcare professional.